Intergenerational Income Mobility under Informal Income and Measurement Error

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1 Informal income as classical measurement error

Let y_{si} and y_{fi} denote the logarithm of the long-run income status of the child and the parent, respectively.

We define the *intergenerational income elasticity* (IGE) as the population correlation between the long-run economic status of sons and their fathers, serving as a measure of intergenerational economic mobility:[1]

$$\rho = \frac{\mathrm{Cov}(y_{si},\, y_{fi})}{\mathrm{Var}(y_{fi})}.$$

Following the standard approach in the literature, we assume a linear relationship between parental and offspring incomes:

$$y_{si} = \alpha + \beta y_{fi} + \varepsilon_i,$$

where $\beta > 0$ represents the causal effect of parental income on the child's long-run income, reflecting the extent to which economic advantages (or disadvantages) are transmitted across generations. A higher value of β indicates lower intergenerational mobility, as it reflects a stronger average causal association between parental income and the long-run income of their children across the population.[2]

Estimating by Ordinary Least Squares (OLS) yields:

$$y_{si} = \hat{\alpha} + \hat{\rho}\, y_{fi} + e_i,$$

where

$$\hat{\rho} = \frac{\widehat{\mathrm{Cov}(y_{si}, y_{fi})}}{\widehat{\mathrm{Var}(y_{fi})}}, \qquad \text{and} \qquad \mathrm{Cov}(y_{fi}, e_i) = 0.$$

[1] Other common measures of intergenerational mobility include the rank–rank regression and transition matrices, as well as the absolute mobility indicators proposed by Chetty, Hendren, Kline, and Saez (2014) and further discussed by Berman (2022), which capture the fraction of children earning higher real incomes than their parents at comparable ages.

[2] The slope parameter β may differ from the population correlation ρ if the error term ε_i is endogenous or correlated with parental income y_{fi}. This could arise, for instance, from measurement error, omitted variables, or life-cycle differences that affect the estimation of long-run income.

By the weak law of large numbers, this estimator converges asymptotically to the IGE i.e $\text{plim}\,\hat{\rho} = \rho$.

Now suppose that the incomes of parents and sons are observed in administrative records:

$$\tilde{y}_{si} = y_{si} + v_i, \qquad \tilde{y}_{fi} = y_{fi} + u_i,$$

where $\tilde{y}_{si}$ and $\tilde{y}_{fi}$ denote the observed incomes of the child and the parent, respectively. Each observation combines the true long-run income status with an additional component that may reflect transitory fluctuations or unobserved informal earnings.

In practice, the true long-run (or permanent) income of individuals is not directly observable. As emphasized by Friedman (1957) in the *Permanent Income Hypothesis*, the distinction between permanent and transitory income is conceptual: annual income data typically combine both components in unknown proportions. While that framework was originally developed to explain consumption behavior over time, subsequent research on intergenerational mobility faced a similar empirical challenge when attempting to infer long-run economic status from available income sources, such as survey data or administrative records.

In particular, Solon (1992) and Zimmerman (1992) considered that the observed income of parents and children might differ from their true long-run status, introducing a *classical measurement error* structure to account for transitory fluctuations or life-cycle variations that distort intergenerational estimates.

However, when the additional component stems from informal income, this interpretation becomes less straightforward. Informal earnings are often linked to systematic individual factors, such as occupational choice, education, or the institutional environment, rather than random noise. Thus, although one may hypothetically treat informal income as a classical measurement error for analytical purposes, such an assumption is unlikely to hold in practice.

Assumption (Classical measurement error)**.** *Under the assumption of classical measurement error, these components are independent of both the true income of each individual and their intergenerational counterpart:*

A1. $\text{Cov}(v_i, u_i) = 0$

A2. $\text{Cov}(y_{si}, v_i) = 0, \quad \text{Cov}(\varepsilon_i, v_i) = 0$

A3. $\text{Cov}(y_{fi}, u_i) = 0, \quad \text{Cov}(\varepsilon_i, u_i) = 0$

Assumption *A2* implies that $\text{Cov}(y_{fi}, v_i) = 0$, and Assumption *A3* implies that $\text{Cov}(y_{si}, u_i) = 0$.

These properties capture the essence of a classical measurement error: the error term is assumed to be purely random noise, unrelated to any systematic component of income. In particular, measurement errors are uncorrelated not only with the true (latent) income they distort, but also with the corresponding measurement errors and true income values of the counterpart in the other generation. Consequently, classical errors do not convey any information about the underlying economic variable or about intergenerational relationships, serving only to add random variance to the observed measures.[3]

[3]Standard formulations in the literature typically impose a zero-mean assumption on measurement errors. This restriction is not required here: allowing for a strictly positive informal income component is consistent with its economic interpretation and does not conflict with the classical measurement error assumptions, which concern independence rather than distributional properties.

Under these conditions, regressing the observed (administrative) income of children on that of their parents yields an estimator $\hat{\rho}_{OLS}$ that asymptotically provides a lower bound for the true IGE:

$$\text{plim}\,\hat{\rho}_{OLS} = \frac{\text{Cov}(\tilde{y}_{si}, \tilde{y}_{fi})}{\text{Var}(\tilde{y}_{fi})} = \frac{\sigma^2_{y_{fi}}}{\sigma^2_{y_{fi}} + \sigma^2_u}\rho = \lambda\rho \ < \ \rho,$$

where $\lambda = \frac{\sigma^2_{y_{fi}}}{\sigma^2_{y_{fi}} + \sigma^2_u} < 1$ is the attenuation factor.

Hence, if informal income is treated as a classical measurement error, the IGE estimated from administrative data is asymptotically a lower bound of the population IGE (ρ) based on long-run economic status.[4] This reasoning aligns with the findings of Solon (1992), Zimmerman (1992), and Björklund and Jäntti (1997), who emphasize that income data typically capture short-run fluctuations rather than the underlying permanent economic status of individuals.

2 Nonclassical measurement error

Interpreting informal income as a classical measurement error is not a reasonable assumption given the nature of this component of income. Unlike transitory shocks, the permanent informal income is unlikely to be independent of the individual's long-run economic status. There are several reasons to expect a systematic relationship between the two: individuals with higher formal earnings may face fewer incentives or opportunities to participate in informal activities, whereas those with lower formal earnings may rely more heavily on them. As a result, the independence assumptions required for the classical measurement error framework (A2 and A3) are unlikely to hold, implying that informal income should instead be regarded as a source of *non-classical measurement error* rather than a purely random, mean-zero disturbance.

We say that an income variable exhibits **non-classical measurement error** in the presence of informal income if the observed income $\tilde{y}$ is related to the true long-run total income y according to

$$\tilde{y} = y - y^I,$$

where y^I denotes the unobserved long-run (or permanent) informal income component satisfying the conditions defined below.[5]

Assumption (Non-classical measurement error)**.**

B1 ***Non-negativity.*** *The informal income component is always non-negative* ($y^I \geq 0$)*, which implies that the observed income systematically understates total income* ($\mathbb{E}[\tilde{y}] < \mathbb{E}[y]$)*. Unlike classical measurement error (mean zero), this error is biased and one-sided.*

[4] If classical measurement error affects only the child's income, while the parent's income is measured without error, the OLS estimator of the IGE remains consistent for the population parameter.

[5] Note that the informal income component refers to the *long-run* or *permanent* informal income, rather than a transitory fluctuation. It is possible to represent both formal and informal long-run incomes as including their respective transitory components, but doing so does not alter the interpretation of classical measurement errors. Within this framework, treating informal income as a non-classical component does not introduce fundamentally different economic or econometric implications beyond those arising from the presence of classical measurement error.

B2 ***Negative correlation between informal and total income (across generations*** *The informal income component is negatively correlated with the long-run (total) income of the counterpart in the other generation, i.e.*

$$\mathrm{Cov}(y^I, y') < 0,$$

where y denotes the income (with an informal component) of one generation and y' the total income of its intergenerational counterpart.

After defining the assumptions, we note that the negative intergenerational correlation (*B2*) is both empirically and theoretically plausible. In an intergenerational context, when the measurement error arises from the child's income, this implies that individuals from high-income families are less likely to rely on informal income in their long-run status, whereas those from lower-income families may face stronger incentives or necessities to engage in informal work.[6]

This assumption is further supported by evidence on the relationship between education and informality. Individuals from higher-income families tend to have greater access to education, which in turn facilitates their transition to and permanence in the formal labor market. As shown by Romanello and Oliveira-Goncalves (2017), educational attainment is the most relevant characteristic explaining workers' exit from informality in Brazil. Therefore, if children from wealthier families achieve higher education levels, they are less likely to depend on informal income sources in their adult life, reinforcing the expected negative intergenerational association between parental income and the informal income share of the offspring.

Similarly, when the measurement error stems from the parent's income, an analogous mechanism operates: parents with a larger informal income component are more likely to have children whose observed long-run status income is relatively lower. Thus, *B2* captures a symmetric form of non-classical measurement error that links informal income and total income across generations.[7]

By explicitly incorporating these two assumptions, our framework captures a realistic form of non-classical measurement error that links informal income to intergenerational differences in total income. This formulation also allows for scenarios in which measurement error affects only certain income variables—such as the child's income—while others, like the father's income, are observed without error. Specifically, the child's observed income is

$$\tilde{y}_{si} = y_{si} - y^I_{si}, \quad y^I_{si} \geq 0,$$

[6] This interpretation should not be confused with the idea that coming from a higher-income family necessarily implies lower earnings in the informal sector. Rather, since we are interested in measures that capture an individual's long-term income, it is implicit that such measures account for the probability of relying on informal income over time. Individuals from wealthier families are much more likely to work in the formal sector and to obtain most of their income from it.

[7] It is worth noting that assumption B2 holds when two separate conditions are satisfied: one directly imposing that the informal income component is negatively correlated with long-run status income, i.e. $\mathrm{Cov}(y^I, y) < 0$, and another requiring that the informal income component be uncorrelated with the idiosyncratic error term in the child's income regression, ($\mathrm{Cov}(y^I, \varepsilon_i) = 0$). The first condition is empirically well supported: individuals with higher total income typically have less incentive or need to engage in informal economic activities, either because they already earn sufficient income through formal channels or because higher-income occupations tend to be more regulated and closely monitored. This pattern has been widely documented in the literature on informal labor markets and income reporting (see, e.g., Schneider, Buehn, and Montenegro 2010; Gomes, Iachan, and Santos 2020), supporting the plausibility of $\mathrm{Cov}(y, y^I) < 0$.

which is subject to non-classical measurement error (assumptions *B1, B2*). Under these assumptions and the specified linear relationship

$$y_{si} = \alpha + \beta y_{fi} + \varepsilon_i,$$

it follows that $\text{Cov}(y^I_{si}, y_{fi}) < 0$. Meanwhile, the father's income is observed without error.

If we calculate the intergenerational elasticity (IGE) using the child's income with non-classical measurement error, the probability limit of the OLS estimator is

$$\text{plim}\, \hat{\rho}_{OLS} = \frac{\text{Cov}(\tilde{y}_{si}, y_{fi})}{\text{Var}(y_{fi})} = \frac{\text{Cov}(y_{si}, y_{fi}) - \text{Cov}(y^I_{si}, y_{fi})}{\text{Var}(y_{fi})} > \rho.$$

Since assumption *B2* holds ($\text{Cov}(y^I_{si}, y_{fi}) < 0$), the asymptotic limit of the IGE estimator becomes an upper bound of the true IGE when the child's long-run status income is latent and only its formal component is available, subject to non-classical measurement error arising from informal income.

The intuition behind obtaining an upper bound lies in the comparison between intergenerational relationships based on total and formal incomes. The child's informal income tends to weaken the overall association, as the relationship between the child's informal income and the parent's total income moves in opposite directions: higher parental income is typically associated with greater educational attainment and other socioeconomic advantages, which, in turn, lead to a smaller share of the child's total income coming from informal sources.

This finding stands in contrast to the asymptotic limit of the IGE under classical measurement error, as well as to the view (as cited in Millimet, Li, and Roychowdhury (2020) from Glewwe (2011)) that "all indices of relative mobility tend to exaggerate mobility when income is measured with error."

2.1 Missing parental income

So far, we considered the scenario where the father's income is observed without error, while the child's income is subject to non-classical measurement error, which leads the OLS estimator to be an upper bound of the true IGE. We now discuss more general scenarios, linking our current framework with cases previously studied in the literature.

Beyond this benchmark case, empirical analyses often face important data limitations. In many applications, researchers rely on administrative sources—such as tax records or unemployment insurance databases—that provide information on formal earnings. In these settings, we typically observe the formal income of individuals (or a proxy, such as the average formal income over several years) for the child generation, but lack information on their long-run informal income. At the same time, data for the parental generation are often more restricted, containing only a few socioeconomic characteristics rather than the parents' true long-run total income.

This situation is particularly common in emerging economies, where long-term intergenerational income data are scarce or fail to provide consistent coverage across generations. Consequently, researchers must impute or estimate parental long-run income to recover a measure of the intergenerational income elasticity (IGE) using the TSTSLS methodology. This framework allows us to assess whether the absence of information on children's informal income leads to inconsistencies in the estimated IGE based on total income, and to identify the conditions under which consistency can still be achieved.

Suppose first that both the child's and the father's long-run incomes are observed without error. In that case, the probability limit of the OLS estimator of the IGE is

$$\text{plim}\, \hat{\rho}_{OLS} = \beta + \frac{\text{Cov}(\varepsilon_i, y_{fi})}{\text{Var}(y_{fi})},$$

where ε_i is the idiosyncratic shock to the child's income.

Next, consider the scenario where the father's income is not observed, but some parental characteristics reported by the child are available. Denote the imputed father's income by $\hat{y}_{fi}$. In this case, the child's observed income is still subject to non-classical measurement error:

$$\tilde{y}_{si} = y_{si} - y_{si}^I, \quad y_{si}^I \geq 0,$$

and the linear relationship is

$$y_{si} = \alpha + \beta y_{fi} + \varepsilon_i.$$

Using two-stage techniques (TSTSLS) with the imputed father's income, the probability limit of the estimator becomes

$$\text{plim}\, \hat{\rho}_{TSTSLS} = \frac{\text{Cov}(\tilde{y}_{si}, \hat{y}_{fi})}{\text{Var}(\hat{y}_{fi})} = \frac{\beta\, \text{Cov}(y_{fi}, \hat{y}_{fi}) + \text{Cov}(\varepsilon_i, \hat{y}_{fi}) - \text{Cov}(y_{si}^I, \hat{y}_{fi})}{\text{Var}(\hat{y}_{fi})}.$$

Assuming that the sample of pseudo-parents is asymptotically equivalent to the true parental sample, the sample covariance between the true and imputed parental incomes coincides with the sample variance of the imputed measure, that is, $\text{Cov}(y_{fi}, \hat{y}_{fi}) = \text{Var}(\hat{y}_{fi})$ (see Appendix A).

This allows us to simplify the expression as:

$$\frac{\text{Cov}(\tilde{y}_{si}, \hat{y}_{fi})}{\text{Var}(\hat{y}_{fi})} = \beta + \frac{\text{Cov}(\varepsilon_i, \hat{y}_{fi}) - \text{Cov}(y_{si}^I, \hat{y}_{fi})}{\text{Var}(\hat{y}_{fi})}.$$

Comparing this to the true IGE ρ, we have

$$\frac{\text{Cov}(\tilde{y}_{si}, \hat{y}_{fi})}{\text{Var}(\hat{y}_{fi})} - \rho = \frac{\text{Cov}(\varepsilon_i, \hat{y}_{fi}) - \text{Cov}(y_{si}^I, \hat{y}_{fi})}{\text{Var}(\hat{y}_{fi})} - \frac{\text{Cov}(\varepsilon_i, y_{fi})}{\text{Var}(y_{fi})}.$$

The TSTSLS estimator is consistent when:[8]

$$\frac{\text{Cov}(\varepsilon_i, \hat{y}_{fi}) - \text{Cov}(y_{si}^I, \hat{y}_{fi})}{\text{Cov}(\varepsilon_i, y_{fi})} = \frac{\text{Var}(\hat{y}_{fi})}{\text{Var}(y_{fi})} = R^2.$$

In contrast to the case in which the father's income is observed and the child's income is affected by non-classical measurement error, so that the OLS estimator systematically overstates the true IGE, in the setting where the father's income is missing and imputed through reported parental characteristics the TSTSLS estimator does not necessarily display an upward bias. Depending on the joint covariance structure of the imputation error, the child's income shock, and the true parental income, the estimator may converge to the true parameter. This shows that imputing parental income, while seemingly adding an additional source of error, may generate scenarios in which the upwards bias is eliminated.

Another relevant aspect to consider is that if we could perfectly predict the true parental income at the population level ($R^2 = 1$) this does not imply that the TSTSLS estimator is consistent. Furthermore, the TSTSLS estimator is inconsistent and becomes an upper bound of the true IGE.

[8] It is important to note that this is not the sample R^2, but rather the stochastic (population) R^2. Additionally, this term represents the population projection coefficient obtained when regressing the predicted income on the actual income, i.e : $\frac{\text{Cov}(\hat{y}_{fi}, y_{fi})}{\text{Var}(y_{fi})}$

It is important to note, however, that this issue does not arise in the absence of informal income—that is, when no non-classical measurement error is present. As Orihuela, Díaz, Cubillos, and Troncoso (2025) point out, when all income is formal, a value of $R^2 = 1$ indeed implies consistency.[9]

Proposition 2.1. *If $R^2 = 1$, the TSTSLS estimator is inconsistent and represents an upper bound of the true IGE.*

Proof. Since the pseudo-parent sample is asymptotically representative of the true parental distribution, it follows that

$$\mathrm{Var}(y_{fi}) = \mathrm{Var}(\hat{y}_{fi}) + \mathrm{Var}(e_i),$$

where $e_i = y_{fi} - \hat{y}_{fi}$ denotes the residual. By definition, $R^2 = \frac{\mathrm{Var}(\hat{y}_{fi})}{\mathrm{Var}(y_{fi})}$, so when $R^2 = 1$ we must have $\mathrm{Var}(e_i) = 0$, which implies $y_{fi} = \hat{y}_{fi}$ for all i. In this case,

$$\mathrm{Cov}(\varepsilon_i, \hat{y}_{fi}) = \mathrm{Cov}(\varepsilon_i, y_{fi}).$$

Assuming, for the sake of contradiction, that the estimator is consistent, and recalling the definition of consistency, we have:

$$\frac{\mathrm{Cov}(\varepsilon_i, \hat{y}_{fi}) - \mathrm{Cov}(y^I_{si}, \hat{y}_{fi})}{\mathrm{Cov}(\varepsilon_i, y_{fi})} = \frac{\mathrm{Var}(\hat{y}_{fi})}{\mathrm{Var}(y_{fi})} = R^2.$$

Then, under the condition $R^2 = 1$, it follows that

$$1 - \frac{\mathrm{Cov}(y^I_{si}, \hat{y}_{fi})}{\mathrm{Cov}(\varepsilon_i, y_{fi})} = 1.$$

However, under assumption *B2*, the presence of non-classical measurement error, we know that $\mathrm{Cov}(y^I_{si}, y_{fi}) < 0$. Consequently,

$$1 - \frac{\mathrm{Cov}(y^I_{si}, \hat{y}_{fi})}{\mathrm{Cov}(\varepsilon_i, y_{fi})} > 1,$$

which contradicts the previous equality.

Furthermore, the TSTSLS estimator acts as an upper bound of the true IGE. Rewriting the difference between the estimated and true parameters, and noting that $R^2 = 1$ and $\mathrm{Cov}(y^I_{si}, y_{fi}) < 0$, it follows that

$$\frac{\mathrm{Cov}(\tilde{y}_{si}, \hat{y}_{fi})}{\mathrm{Var}(\hat{y}_{fi})} - \rho = -\frac{\mathrm{Cov}(y^I_{si}, y_{fi})}{\mathrm{Var}(\hat{y}_{fi})} > 0.$$

□

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A Asymptotic Properties of Predicted Parental Income

Proposition A.1 (Sample covariance and variance of $\hat{y}_{fi}$)**.** *Let $\hat{y}_{fi} = X_i'\hat{\delta}$, where $\hat{\delta}$ is an OLS estimator obtained from a sample of pseudo parents asymptotically equivalent to the real parent sample. Then:*

$$\operatorname{plim} \widehat{\operatorname{Cov}(y_{fi}, \hat{y}_{fi})} = \operatorname{plim} \widehat{\operatorname{Var}(\hat{y}_{fi})} = \operatorname{Cov}(y_{fi}, X_i) \operatorname{Var}(X_i)^{-1} \operatorname{Cov}(X_i, y_{fi}).$$

Proof. By definition of sample covariance and variance:

$$\widehat{\operatorname{Cov}(y_{fi}, \hat{y}_{fi})} = \hat{\delta}' \widehat{\operatorname{Cov}(y_{fi}, X_i)}, \qquad \widehat{\operatorname{Var}(\hat{y}_{fi})} = \hat{\delta}' \widehat{\operatorname{Var}(X_i)} \hat{\delta}.$$

Applying the probability limit and using that the pseudo-parent sample is asymptotically equivalent to the real parent sample, we have

$$\operatorname{plim} \hat{\delta} = \operatorname{Var}(X_i)^{-1} \operatorname{Cov}(X_i, y_{fi}).$$

Therefore:

$$\operatorname{plim} \widehat{\operatorname{Cov}(y_{fi}, \hat{y}_{fi})} = (\operatorname{plim} \hat{\delta})' \operatorname{Cov}(y_{fi}, X_i) = \operatorname{Cov}(y_{fi}, X_i) \operatorname{Var}(X_i)^{-1} \operatorname{Cov}(X_i, y_{fi}),$$

and similarly:

$$\operatorname{plim} \widehat{\operatorname{Var}(\hat{y}_{fi})} = (\operatorname{plim} \hat{\delta})' \operatorname{Var}(X_i) (\operatorname{plim} \hat{\delta}) = \operatorname{Cov}(y_{fi}, X_i) \operatorname{Var}(X_i)^{-1} \operatorname{Cov}(X_i, y_{fi}).$$

Hence, the equality of probability limits holds $\operatorname{plim} \widehat{\operatorname{Cov}(y_{fi}, \hat{y}_{fi})} = \operatorname{plim} \widehat{\operatorname{Var}(\hat{y}_{fi})}$. □